Microbes in Human welfare

Introduction and Microbes in Household Products

1. Read the following four statements. Find out the correct option with respect to lactic acid bacteria.

A. They produce acids that coagulate and partially digest milk fat.
B. They require suitable temperature for their multiplication.
C. They play a beneficial role by checking disease causing microbes in our stomach.
D. They improve the nutritional quality by increasing the amount of riboflavin.

1) (A) and (C)
2) (B) and (D)
3) (B) and (C)
4) (A), (C), and (D)

2. In the fermentation of dough and cheese making, the main gas produced is

1) Methane and H_2
2) Methane, CO_2, and H_2
3) Carbon dioxide
4) Carbon monoxide

3. The vitamin whose content increases following the conversion of milk into curd by lactic acid bacteria is

1) Vitamin C
2) Vitamin D
3) Vitamin B_{12}
4) Vitamin E

4. The microscopic proteinaceous infectious agents are called

1) Virus
2) Viroids
3) Prions
4) Virusoids

5. Complete the following analogy:
Swiss cheese: Propionibacterium sharmanii : Bread : _____

1) Candida albicans
2) Lactic acid bacteria
3) Clostridium butylicuит
4) Saccharomyces cerevisiae.

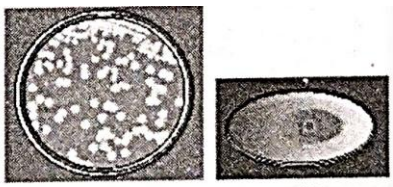
6. Microorganisms are present in

1) Soil, water, and air
2) The bodies of plants and animals
3) Thermal vents
4) All of the above

7. Inoculum is added to milk during curd formation in order to

1) Increase the vitamin B_2 content
2) Coagulate the milk proteins
3) Improve the flavor of milk
4) Partially digest the milk fat

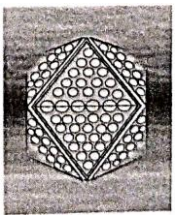
8. Identify the figures and find out the correct option.



(A) (B)

1) A-Bacterial colony; B-Fungal colony
2) A-Fungal colony; B-Bacterial colony
3) A-Phage colony; B-Bacterial colony
4) A-Bacterial colony; B-Phage colony

9. Look at the figure and select the correct option.



1) This is a prokaryote that causes respiratory disorder.
2) This is a nucleoprotein particle that is capable of causing urinary infections.
3) This is an adenovirus that causes respiratory infections.
4) This is a fungal colony growing in a Petri dish.

Microbes in Industrial Products

10. The chemical substances produced by some microbes that can kill or retard the growth of other microbes are called
1) Antibiotics
2) Probiotics
3) Bioactive molecules
4) Prebiotics

11. The large vessels for growing microbes on an industrial scale are called
1) Tanks
2) Digesters
3) Fermentors
4) All of the above

12. Microbes are also used in the commercial production of certain organic acids. Butyric acid is produced by _(A)_, and lactic acid is produced by_ (B)_.
1) A-Clostridium butylicum; B-Lactobacillus
2) A-Lactobacillus; B-Aspergillus
3) A-Clostridium butylicum; B—Aspergillus
4) A-Acetobacter aceti; B-Aspergillus

13. Which one of the following alcoholic drinks is produced without distillation?
1) Wine
2) Whisky
3) Rum
4) Brandy

14. Read the given assertion and reason carefully and select the correct option.
Assertion: Ethanol is produced by fermenting malted cereals and fruit juices by yeast.
Reason: Wine and beer are produced by distillation of the fermented broth.

1) If both assertion and reason are true and the reason is the correct explanation of the assertion.
2) If both assertion and reason are true, but reason is not the correct explanation of the assertion.
3) If assertion is true, but reason is false.
4) If both assertion and reason are false.

15. Read the following statements with respect to antibiotics, and state true (T) or false (F).

A. Antibiotics means "prolife" in the context of diseasecausing organisms.
B. Waksman discovered Streptomycin and coined the term antibiotics.
C. Full potential of penicillin as an effective antibiotic was established by Chain and Florey.
D. Penicillin was extensively used to treat American soldiers wounded in World War II.

1) A—T; B—T; C-F; D-F
2) A-F; B-T; C-T; D-T
3) A-F; B—T, C—F; D-F
4) A-T; B-F; C-T; D-T

16. Read the following statements and choose the correct option.

1) Yeast Saccharomyces cerevisiae is used for breadmaking and beer making.
2) Microbes especially yeasts have been used from time immemorial for the production of beverages like wine, beer, whisky, brandy or rum.
3) After penicillin, other antibiotics were also purified from other microbes.
4) All of the above

17. Antibiotics have greatly improved our capacity to treat the following deadly disease except

1) Plague
2) Diphtheria
3) Kali khansi
4) Cancer

18. Which drugs are administered before and after an organ transplant procedure?

1) Streptokinase
2) Cyclosporin A
3) Anabolic steroids
4) All of the above

19. A patient brought to a hospital with myocardial infarction is immediately given

1) Penicillin
2) Statins
3) Streptokinase
4) Cyclosporin A

Microbes in Sewage Treatment and Microbes in Production of Biogas

20. Wastewater treatment generates a large quantity of sludge, which can be treated by

1) Anaerobic digesters
2) Floc
3) Chemicals
4) Oxidation pond

21. The microorganisms involved in floc formation during sewage treatment are

1) Aerobic bacteria and fungi
2) Anaerobic bacteria and fungi
3) Autotrophic bacteria and fungi
4) All of the above

22. Read the following statements and select the correct option.

Statement A: A BOD is the measure of amount of O_2 that would be consumed if all the inorganic and organic matter in 1 L water is oxidized by bacteria.

Statement B: The greater the BOD of wastewater, the lesser is its pollution potential.

1) Only statement A is correct
2) Only statement B is correct
3) Both statements A and B are correct
4) Both statements A and B are incorrect

23. Given below are three statements (i-iii) each with one blank. Select the correct option that fills the blanks in the statement.

(i) Formation of flocs is observed during A.

(ii) The supernatant left after the sedimentation of sewage water during secondary treatment is known as _B_.

(iii) BOD is the measure of C in wastewater.

1) A-Biological treatment; B-Activated sludge; C-Organic load
2) A-Secondary treatment; B--Secondary effluent; C-Organic load
3) A-Biological treatment; B-Anaerobic sludge; C-Inorganic load
4) A-Secondary treatment; B-Primary sludge; C-Flocs

24. State true (T) or false (F) for the following statements with respect to sewage treatment.

A. The treatment of waste water in a sewage treatment plant is done by autotrophic microorganisms present in the sewage.

B. During secondary treatment in a secondary treatment plant, BOD initially increases and then decreases.

C. Anaerobic digestion of activated sludge produces biogas inside the sludge digester.

D. A small portion of primary sludge is used as an inoculum to start another secondary treatment.

1) A-F; B-T; C—T; D-F
2) A-T; B—T; C—F; D-T
3) A-F; B—F; C—T; D-F
4) A-T; B—F; C—T; D-F

25. The Ministry of Environment and Forest has initiated Ganga and Yamuna Action Plans to

1) Treat wastewater
2) Prevent the discharge of untreated sewage into these rivers
3) Prevent microbial treatment of sewage
4) Both (1) and (2)

26. Pick the incorrect match regarding sewage.

1)	High BOD of water	More polluted water
2)	Activated sludge	Serves as inoculum in aeration tank
3)	Anaerobic sludge digester	Aerobic flocs
4)	Physicochemical process	Tertiary treatment

27. The primary treatment of sewage involves

1) Decomposition
2) Microbial activity
3) Sequential filtration and sedimentation
4) Fermentation

28. Read the following statements.

Statement A: Activated sludge in settlement tanks of sewage treatment plant is a rich source of aerobic bacteria.

Statement B: Biogas is produced by the activity of aerobic bacteria on animal waste (dung).

Choose the correct option.

1) Statement B is correct and statement A is wrong.
2) Both the statements A and B are correct.
3) Statement A is correct and statement B is wrong.
4) Both the statements A and B are wrong

29. Read the following statements with respect to biogas production, and state true (T) and false (F).

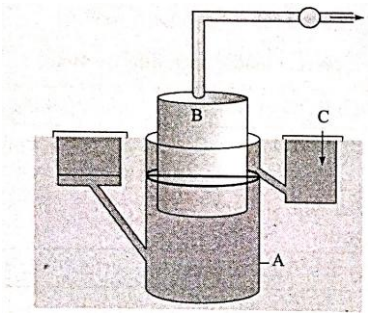
A. The type of the gas produced depends upon the microbes and the organic substrate they utilize.
B. Certain bacteria that grow aerobically on cellulose material produce large amount of methane along with CO_2 and H_2.
C. Methanobacterium is a common methanogen.
D. Methanogens are commonly found in the anaerobic sludge during sewage treatment.

1) A-T; B-F; C-T; D-F
2) A-F; B—T; C—T; D-T
3) A-T; B-F; C-T; D-T
4) A-F; B-F; C-T; D-T

30. Choose the incorrect statement with respect to biogas.

1) It predominantly contains methane.
2) Its production involves the aerobic digestion of organic waste
3) It is also called gobar gas.
4) Methanobacterium is a common biogas-producing microbe.

31. The figure given below shows a typical biogas plant. In which of the following option, products labelled as A, B, and C are correctly identified?



1) A—Sludge; B—CH_4 + CO_2; C-Dung + water
2) A-Digester; B-CH_4+ CO_2; C-Sludge
3) A-Sludge; B-CH_4 + NO_2; C-Sewage
4) A-Digester; B-CH_4 + CO_2; C-Dung+ water

32. Methanogens do not produce

1) Oxygen
2) Methane
3) Hydrogen sulfide
4) Carbon dioxide

33. Which of the following term is not associated with typical biogas plant?

1) Concrete tank
2) Floating cover
3) Outlet
4) Trickling filter

34. Find out which of the following statement is incorrect with respect to biogas plant.

1) It has a floating cover that keeps on rising as gas is produced.

2) It consists of a 10 to 15-foot deep tank in which biowastes are collected and a slurry of dung is fed.
3) Spent slurry may be used as fertilizers.
4) Main gases produced are methane, CO and nitrogen oxide

35. Read the following statements w.r.t. activated sludge and select the correct option:
A. After secondary treatment, microbial flocs are allowed to settle down, which is called as activated sludge.
B. The major part of the sludge is pumped into large tanks called anaerobic sludge digester.
C. It is generally released into natural water bodies like rivers and streams.
1) Only (A) is correct
2) Only (A) and (C) are correct
3) Only (A) and (B) are correct
4) Only (B) is correct

Microbes as Biocontrol Agents and Microbes as Biofertilizers

36. Which of the following organisms is not used during biocontrol method?
1) Bacillus thuringiensis
2) Trichoderma
3) Ladybird beetle
4) Monascus purpureus

37. The Nucleopolyhedrovirus (NPV) are
1) Not baculoviruses
2) Effective against rodents
3) Broad-spectrum insecticides
4) Effective against insects and other arthropods

38. Which of the following is incorrect about Baculoviruses?
1) Pathogens that attack insects and other arthropods
2) Species-specific insecticidal applications
3) Broad spectrum insecticidal applications
4) No negative impacts on plants, mammals, birds, fish or even on non-target insects

39. Which larvicidal fish is used to control the mosquito larvae?
1) Gambusia
2) Singhara fish
3) Dog fish
4) Electric ray

40. Which of the following is an example of carrying out biological control of pest or disease-causing microbes?
1) Bt spores to increase sugarcane yield.
2) Nucleopolyhedrovirus against mammals, birds, and fish.
3) Ladybird beetle against mosquito
4) Trichoderma species against certain plant pathogens.

41. Go through the following matches with respect to biocontrol agents, and select the correct option.
A. Ladybird beetle-Aphids
B. Dragonflies-Mosquitoes
C. Bacillus thuringiensis-Butterfly caterpillars
D. Trichoderma-Plant pathogens

1) Only (A) and (B) are correct.
2) Only (C) is correct.
3) Only (D) is correct.
4) All (A), (B) (C) and (D) are correct.

42. Match the items in column 'A' and column 'B' and choose the correct answer.

	Column A		**Column B**
(i)	Ladybird	(A)	Methanobacteriuт
(ii)	Mycorrhiza	(B)	Trichoderma
(iii)	Biological control	(C)	Aphids
(iv)	Biogas	(D)	Glomus

1) i-B, ii-D, iii-C, iv-A
2) i-C, ii-D, iii-B, iv-A
3) i-D, ii-A, iii-B, iv-C
4) i-C, ii—B, iii—A, iv-D

43. Identify the correct statement.

A. Baculoviruses are pathogens that attack insects and other arthropods.
B. Nucleopolyhedroviruses are excellent candidates for species-specific narrow-spectrum insecticidal application.
C. Bacillus thuringiensis are available in sachets as dried spores which are mixed with organic solvent and sprayed onto vulnerable plants.
D. Trichoderma species are free-living fungi that are very common in the root ecosystem.

1) Only (A) and (B) are correct.
2) Only (A) and (C) are correct.
3) (B), (C) and (D) are correct.
4) (A), (B) and (D) are correct.

44. Microbes used to control pests under IPM include

1) Dragonfly
2) Ladybird
3) Trichoderma
4) Cactoblastis

45. Consider the following four statements (A-D) related to organic farming and select the correct option stating which ones are true (T) and false (F).

A. The pests are not eradicated but instead are kept at manageable levels by a simple system of checks and balances with in a living and vibrant ecosystem.
B. The more variety a landscape has, the less sustainable it is.
C. Bioinsecticides and biopesticides are much better option than chemical insecticides and pesticides.
D. Organic farming involves the use of only chemical fertilizers.

1) A-F; B-F; C-T; D-F
2) A-T; B-F; C-T; D-F
3) A-F; B-T; C-T; D-F
4) A-F; B-T; C-F; D-T

46. Read the following statements and choose the correct option.

Statement 1: Bacillus thuringiensis are available in sachets as dried spores which are mixed with water and sprayed onto vulnerable plants such as Brassica and fruit trees, where these are eaten by the insect larvae. In the gut of the larvae, the toxin is released, and the larvae get killed.

Statement 2: The bacterial disease will kill the caterpillars but leave other insects unharmed.

1) Both statements are correct.

2) Both statements are incorrect.
3) Only statement 1 is correct but statement 2 is incorrect.
4) Only statement 2 is correct but statement 1 is incorrect.

47. An organism used as biofertilizer for raising soyabean crop is
1) Azospirillum
2) Rhizobium
3) Frankia
4) Anabaena

48. Biofertilizers are microorganisms that bring about
1) Soil nutrient enrichment
2) Minimizing ecological benefits
3) Maximizing environmental hazards
4) All of the above

49. Which one of the following is not a nitrogen-fixing organism?
1) Anabaena
2) Nostocc
3) Azotobacter
4) Pseudomonas

50. Read the following statements and choose the correct option.
I. In non-leguminous plants, Rhizobium fix atmospheric nitrogen into inorganic forms, which is used by the plant as nutrient.
II. The fungal symbiont in mycorrhiza absorbs phosphorus from soil and passes it to the plant.
III. There are problems associated with the overuse of chemical fertilisers and there is a large pressure to switch to inorganic farmingthe use of chemical fertilisers.
1) All statements are correct
2) All statements are correct except I
3) All statements are incorrect except II
4) All statements are correct except III

51. Which of the following are free-living nitrogen-fixing bacteria found in soil?
1) Rhizobium and Frankia
2) Azolla and Nostoc
3) Azospirillum and Azotobacter
4) Frankia and Oscillatoria

52. Biological control of pest is
1) Polluting
2) Highly expensive
3) Self-perpetuating
4) Toxic

Answer Key

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**
3	3	3	3	4	4	2	1	3	1	3	1	1	3	2
16	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
4	4	2	3	1	1	4	2	1	4	3	3	3	3	2
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**	**41**	**42**	**43**	**44**	**45**
2	1	4	4	3	4	4	3	1	4	4	2	4	3	2
46	**47**	**48**	**49**	**50**	**51**	**52**								
1	2	1	4	3	3	3								